

Lecture 08 SQL WorkSheet

Create three tables Emp , Dept and Location to store details of Employees and Departments with below fields.

Emp (empId , fName , lName ,email ,hireMonth,salary , deptNo)

Dept (deptNo , dName , dLoc , createDate, DepManagerName)

Location (LocID, LocTown, LocProvince)

Employees should be worked in departments which is only stated in Department table. But there can be some employees who are not assigned to any department. Also Department table will have DLocs which is stated under LocID property in location table.

Use below scripts to create and Insert data to above tables

```
CREATE TABLE Location
(
    LocID INT(5),
    LocTown VARCHAR (25),
    LocProvince VARCHAR (25),
    CONSTRAINT l_lno_pk PRIMARY KEY (LocID)
);
```

```
CREATE TABLE Dept
(
    deptNo INT(4),
    dName VARCHAR(20) UNIQUE NOT NULL,
    dLoc INT(5) ,
    createDate TIMESTAMP,
    DepManagerName VARCHAR(25),
    CONSTRAINT d_dno_pk PRIMARY KEY (deptNo),
    CONSTRAINT d_dloc_fk FOREIGN KEY (dLoc) REFERENCES
    Location(LocID)
);
```

```

CREATE TABLE Emp
(
empId INT(6),
fName VARCHAR(50) NOT NULL,
lName VARCHAR(50) NOT NULL,
email VARCHAR(100) UNIQUE NOT NULL,
hireMonth VARCHAR(10),
salary INT,
deptNo INT(4),
CONSTRAINT e_eid_pk PRIMARY KEY (empId),
CONSTRAINT e_dno_fk FOREIGN KEY (deptNo) REFERENCES Dept(deptNo)
);

```

-- Insert Below records to Dept Employee and Location tables

```

INSERT INTO Location (LocID,LocTown,LocProvince)
VALUES
    (100,'Colombo','Western'),
    (200,'Kandy','Central'),
    (300,'Jaffna','Notheren'),
    (400,'Ampara','Eastern'),
    (500, 'Galle','Southern');

```

```

INSERT INTO
Dept (deptNo, dName,dLoc,createDate,DepManagerName)
VALUES
    (10, 'IT', '100','2015-03-08','Mike'),
    (20, 'Operations', '100','2014-03-08','Simon'),
    (30, 'Sales', '400','2015-03-08','Donald'),
    (40, 'Marketing', '200','2016-03-08','Joe'),
    (50, 'Management', '200','2016-03-08','Jim'),
    (60, 'Exams', '300','2017-03-08','Elon'),
    (70, 'Security', '400','2018-05-08','Mike'),
    (80, 'Registry', '300','2018-03-08','Sam');

```

```

INSERT INTO
Emp (empId,fName ,lName,email,hireMonth,salary,deptNo)
VALUES
(1001, 'Jim', 'King', 'jk@firm.com', 'January', 25000, 50),
(1002, 'Jane', 'Queen', 'jq@firm.com', 'February', 35000 , 50),
(1003, 'Jen', 'Probert', 'jp@firm.com', 'January', 50000, 10),
(1004, 'Mike', 'Brent', 'mb@firm.com', 'February', 85000 , 10),
(1005, 'Nadia', 'Tamsa', 'nt@firm.com', 'January', 55000, 20),
(1006, 'Lesley', 'Morgan', 'lm@firm.com', 'February', 90000,
30),
(1007, 'Simon', 'King', 'sk@firm.com', 'January', 75000, 20),
(1008, 'Adam', 'Brent', 'ab@firm.com', 'March', 70000, 30),
(1009, 'Russel', 'Brent', 'rb@firm.com', 'March', 70000, 10),
(1010, 'Paul', 'Brent', 'pb@firm.com', 'March', 70000, 20),
(1011, 'Kyle', 'Ewans', 'ke@firm.com', 'March', 70000, 10);

```

Write down SQL statements for below questions

1. Display employee details who takes the highest and lowest salaries in the company
2. Display the number of employees who works in Department 30
3. Display total number of employees who are assigned to a department
4. How many departments are there currently employees are assigned to?
5. Display the Average Salary of an employee who works in Department 10
6. Display the total amount of employee salary allocated for the departments 10 and 20
7. Display department name and the department wise average salary under the column Name "DeptAvg".
8. Find out which department gets the Highest Average Salary
9. Display the number of employees joined to company in each month. Output should contain the name of the month and employees joined under the column named "Total".
10. Display department wise sum of the salary according to the month employee have joined
11. Display the number of employees according to the location they work. Output must include the number, Province and employee count.
12. Display the months where more than two employees have joined to the company
13. List down the names of all departments, and the average salary count where employees earn average salary of more than 50000
14. List down the department names where an employee's highest salary is more than 50000. Do not consider the employees joined in January. Display the final result according to the ascending order of highest salary.
15. Identify the list of towns where more than 3 employees are working except Kandy.
16. Identify the department wise average salary of the employees who joined in January and February. Display departments which has an average salary between 60000 to 70000.